

Università degli Studi di Padova

Messa a punto di un potenziale interatomico per la simulazione della tantala amorfa tramite machine-learning

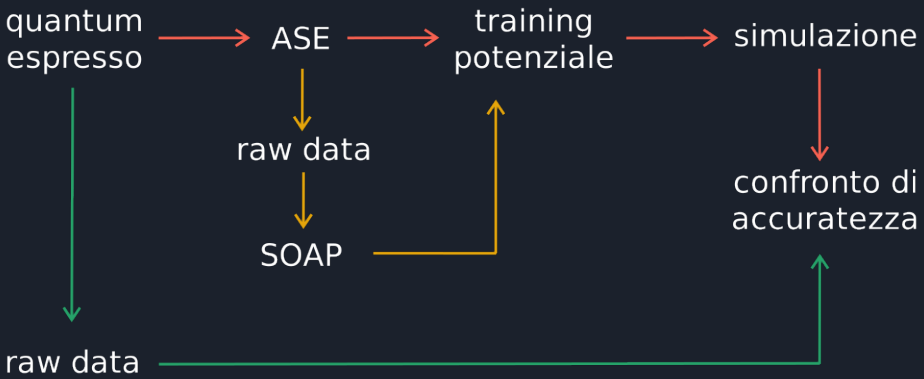
Relatore: Prof. Paolo Umari Laureando: Alberto Stocco

Anno Accademico 2025/2026

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   - Algoritmo Metropolis

Struttura di lavoro

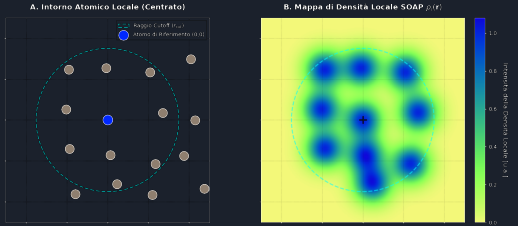


SOAP

SOAP: Smooth Overlap of Atomic Positions

Generazione della mappa di densità

$$\rho_Z(\vec{r}) = \sum_{\alpha}^{Z} e^{-\frac{|\vec{r}-\vec{r}_\alpha|^2}{2\sigma^2}} \qquad (1)$$



SOAP

Espansione in funzioni di base

$$\rho_Z(\vec{r}) = \sum_{n=1}^{n_{max}} \sum_{l=0}^{l_{max}} \sum_{m=-l}^{l} c_Z(n,l,m) g(r\,;n) Y(\theta,\phi\,;l,m) \qquad (2)$$

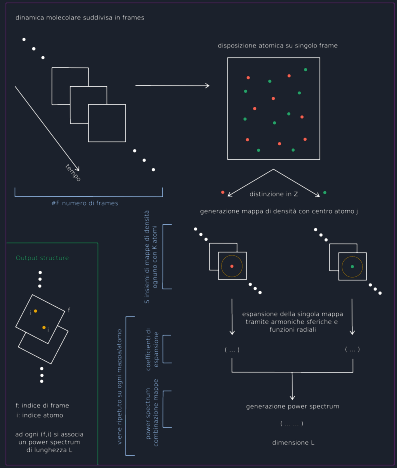
$$c_Z(n,l,m) = \int_{\mathbb{R}^3} \mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi \;\; g(r\,;n) Y(\theta,\phi\,;l,m) \rho_Z(\vec{r}) \qquad (3)$$

Calcolo del power spectrum

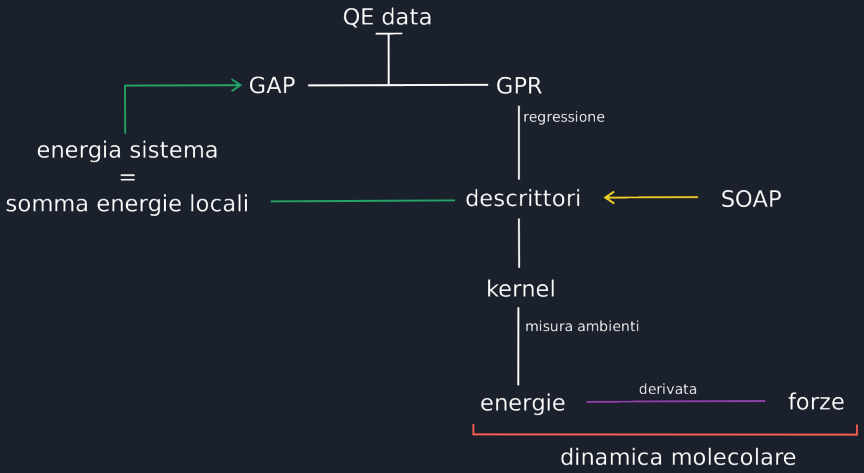
$$p_{Z_1 Z_2}(n,n',l) = \pi \sqrt{\frac{8}{2l+1}} \sum_{m=-l}^{l} \overline{c_{Z_1}}(n,l,m) c_{Z_2}(n',l,m) \qquad (4)$$

Risultato finale

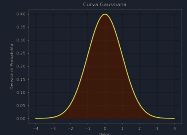
$$\dim = (l_{max}+1)\,\frac{1}{2}\,(S \cdot n_{max})(S \cdot n_{max}+1) \qquad (5)$$



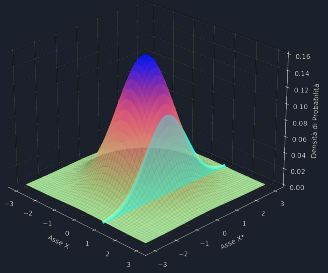
GAP


QE data
GAP
GPR
regressione
energia sistema
=
somma energie locali
descrittori
SOAP
kernel
misura ambienti
energie
derivata
forze
dinamica molecolare

GPR



Gaussiana bivariata



Simulazione NVE

Simulazione dove si impone la conservazione di:

- numero di particelle
- volume
- energia

ZBL

ZBL: Potenziale Ziegler-Biersack-Littmark

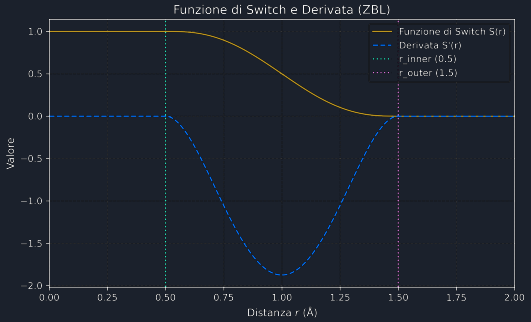
$$V(r_{ij}) = \frac{1}{4\pi\epsilon_0}\frac{Z_i Z_j}{r_{ij}}\phi\left(\frac{r_{ij}}{a_{ij}}\right) \tag{6}$$

$$\phi\left(\frac{r_{ij}}{a_{ij}}\right) = \phi(x) = 0.1818e^{-3.2x} + 0.5099e^{-0.9423x} + 0.2802e^{-0.4029x} + 0.02817e^{-0.2016x} \tag{7}$$

$$a_{ij} = \frac{0.8854\, a_0}{Z_i^{0.23} + Z_j^{0.23}} \tag{8}$$

$$S(u) = -6u^5 + 15u^4 - 10u^3 + 1 \tag{9}$$

$$u = \frac{r - r_{inner}}{r_{outer} - r_{inner}} \tag{10}$$

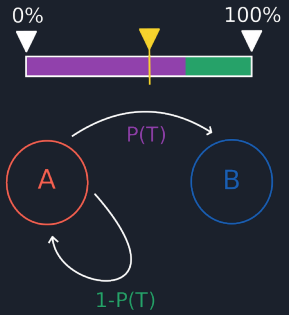


Simulazione con algoritmo di Metropolis

$$P(A) \propto \exp\left(-\frac{E_{pot}(A)}{k_b\,T}\right) \tag{11}$$

$$\Delta E = E_{new} - E_{old} \tag{12}$$

$$P_{acc} = \exp\left(-\frac{\Delta E}{k_b\,T}\right) \tag{13}$$



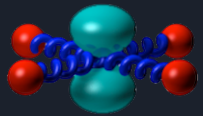
Python

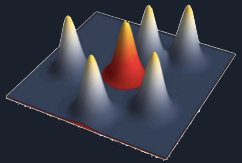
L'analisi è stata svolta in Python utilizzando le seguenti librerie:

- ASE: gestione computazionale della struttura fisica
- DScribe: test del metodo SOAP sui dati
- Quippy: addestramento del potenziale







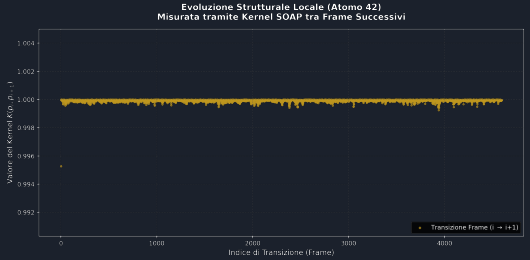


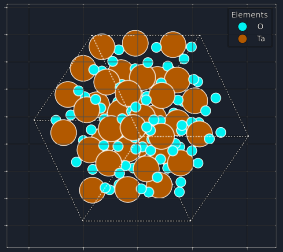
Dataset A

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Divisione dei dati: Training set - Test set

Verifica dei risultati soap



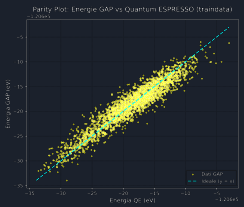


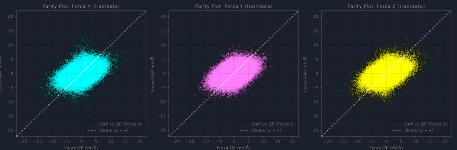
Frame del dataset A generato con matplotlib

Verifica delle energie e delle forze

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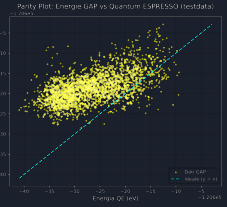
Training set

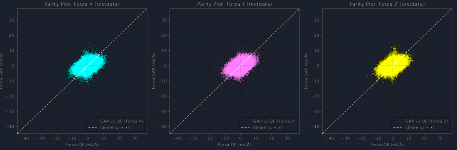




Energie e forze divise per componenti

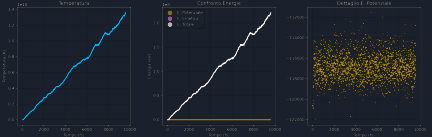
Test set



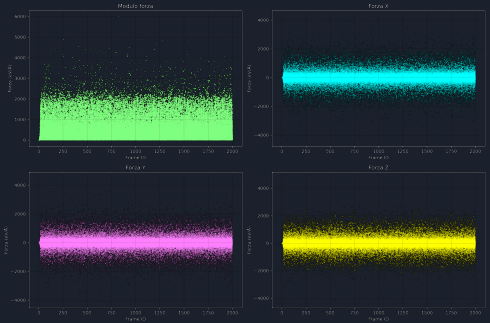


Energie e forze divise per componenti

Simulazione NVE



Andamento temporale di temperatura ed energia

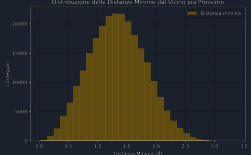


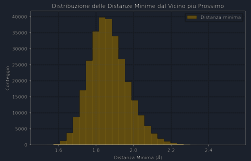
Analisi delle forze, divise per componenti e totale

Simulazione NVE: analisi

Distanza tra i primi vicini

Set di dati simulato con GAP

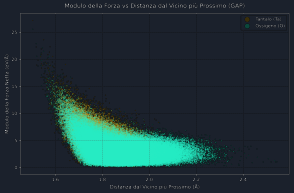




Set di dati di training

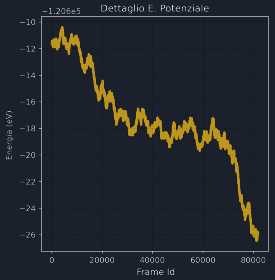
Relazione forza-distanza



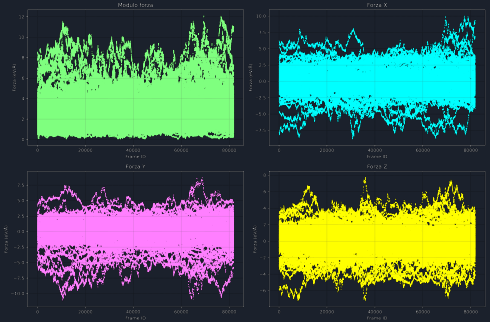


ZBL: risultati

Algoritmo di Metropolis: risultati

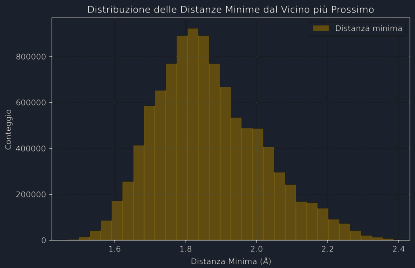


Andamento temporale dei valori di energia

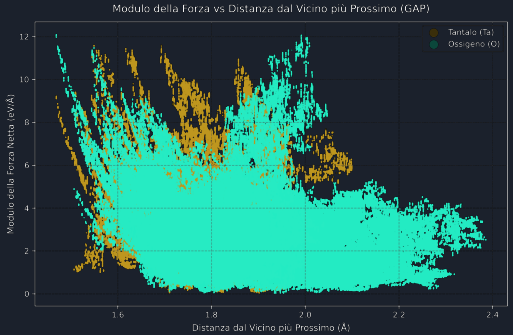


Analisi delle forze, per componenti e totale

Algoritmo di Metropolis: risultati



Distanza tra i primi vicini



Relazione forze-distanza

Domande??